

**REAL TIME PROJECT PRESENTATION
ON Hyperfit AI:your
smart,adaptive and always evolving personal trainer**

23R21A6688-G.PURIJAGANNATH

Under the guidance of

MR P.BABU

Assistant Professor

Department of Computer Science & Engineering-Artificial Intelligence and Machine Learning

OVERVIEW

- Abstract
- Introduction
- Existing Systems
- Challenges Of Existing Systems
- Literature Survey
- Proposed System
- Objectives
- System Requirements
- System Architecture
- UML Diagrams
- Implementation(Sample code)
- Output
- Conclusion
- Future Enhancements
- References

ABSTRACT

A Personal AI Gym Trainer uses AI, computer vision, and machine learning to deliver realtime, personalised fitness coaching. It analyzes movements via pose estimation to ensure proper form and prevent injuries. NLP enables voice interaction, while AI tailors workout plans based on progress and goals. Cloud storage allows easy access across devices, with mobile/web apps for tracking. Advanced analytics offer performance insights, and optional AR/VR adds immersive experiences. This smart trainer makes professional guidance more accessible, affordable, and effective for all fitness levels.

INTRODUCTION

- **WHAT IS HYPERFIT AI ?**

- Advanced AI systems optimized for peak performance and efficiency.
- Designed for specific tasks, industries or real-time applications
- combines cutting-edge algorithms, adaptability, and scalability.



• WHAT IS HYPERFIT AI ?

- Advanced AI systems optimized for peak performance and efficiency.
- Designed for specific tasks, industries or real-time applications
- combines cutting-edge algorithms, adaptability, and scalability.



EXISTING SYSTEMS OF HYPERFIT AI 1. Fitbit & Google Fit

Functionality: Track steps, heart rate, sleep, and calories using wearable sensors. **AI Use:** Offers insights and health trends using simple data analytics.

2. MyFitnessPal

Functionality: Calorie tracking, diet planning, and weight management. **AI Use:** Uses historical data to make food recommendations.

3. Freeletics & Nike Training Club

Functionality: Provide guided workouts with visual coaching. **AI Use:** Adaptive workout plans based on user feedback and progress.

4. Cure.fit (Cult.fit)

Functionality: Offers digital and in-person fitness training, mental wellness, and diet planning.

CHALLENGES OF EXISTING SYSTEMS

- **Limited Personalization**

Most current apps offer generic plans or basic customization.

Lack of deep AI-driven adaptation based on user performance, progress, and preferences.

- **No Real-Time Feedback**

Systems like Fitbit, MyFitnessPal, and Freeletics don't provide instant feedback on exercise form or breathing techniques.

Risk of injury due to incorrect posture or movements goes unnoticed.

- **Manual Data Entry**

Nutrition and activity tracking apps often require manual input, which can be time-consuming and prone to errors or skipped entries.

- **Fragmented Ecosystems**

Users often need multiple apps for fitness, diet, sleep, and mental health.

No unified platform providing a holistic wellness solution.

LITERATURE SURVEY

S.No	Authors/Source	Title	Journal/Publication	Year
1	TechCrunch	<u>Hyperfit</u> AI: The Future of Smart Fitness Training	https://techcrunch.com/hyperfit-ai-future-smart-fitness	2024
2	Men's Health	Train Smarter, Not Harder with Hyperfit AI	https://menshealth.com/train-smarter-hyperfit-ai	2018
3	Forbes	AI-Powered Fitness: <u>Hyperfit</u> AI Targets the Digital Generation	https://forbes.com/ai-powered-fitness-hyperfit-digital	2023
4	The New York Times	Virtual Workouts Go High-Tech with <u>Hyperfit</u> AI	https://nytimes.com/virtual-workouts-hyperfit-ai	2019

PROPOSED SYSTEM

The increasing demand for health, fitness, and home-based exercise routines has created a need for intelligent systems that can monitor and guide users without the presence of a human trainer. Traditional fitness methods often lack real-time feedback, which may lead to incorrect postures and potential injuries.

Moreover, access to professional coaching or physiotherapy may not always be affordable or available. The proposed HyperFit AI system addresses these challenges by offering a cost-effective, realtime pose detection solution using readily available hardware like a webcam and open-source libraries like MediaPipe and OpenCV. These tools enable accurate detection of body landmarks and postural alignment with minimal computational resources.

OBJECTIVES ◦ Personalized Workout Plans

AI analyzes fitness goals, body metrics, and performance data.
Generates tailored exercise routines for optimal results.

◦Real-Time Performance Tracking

Wearable devices with AI monitor heart rate, calories burned. Provides instant feedback and adjustments during workouts.

◦Nutrition Guidance

Creates customized meal plans based on dietary needs & fitness goals Tracks macronutrient intake and suggests adjustments.

◦ Injury Prevention

Analyzes movement patterns to detect improper form or risk of injury.



System Requirements HARDWARE

REQUIREMENTS

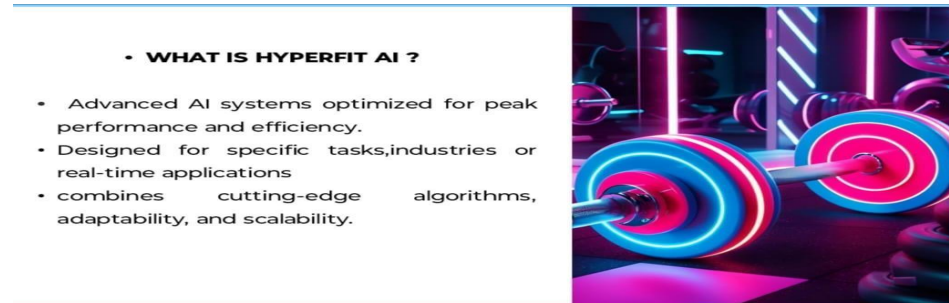
Development Stage:

- CPU: Intel Core i5 or AMD Ryzen 5
(Recommended: Intel Core i7 or AMD Ryzen 7)
- RAM: Minimum 8 GB
(Recommended: 16 GB or more for smoother real-time processing)

- Storage: 256 GB SSD

(Faster read/write speeds for image and model processing)

- GPU (Optional): For future AI enhancements, NVIDIA GeForce GTX/RTX series with CUDA support



System Requirements

SOFTWARE REQUIREMENTS:

Frontend: React.js, Vite, Tailwind CSS

Backend Framework: Express.js, Node.js, TypeScript

Database: Firebase (Authentication + Firestore)

Development Tools: VS Code, GitHub, Cursor

• WHAT IS HYPERFIT AI ?

- Advanced AI systems optimized for peak performance and efficiency.
- Designed for specific tasks, industries or real-time applications
- combines cutting-edge algorithms, adaptability, and scalability.



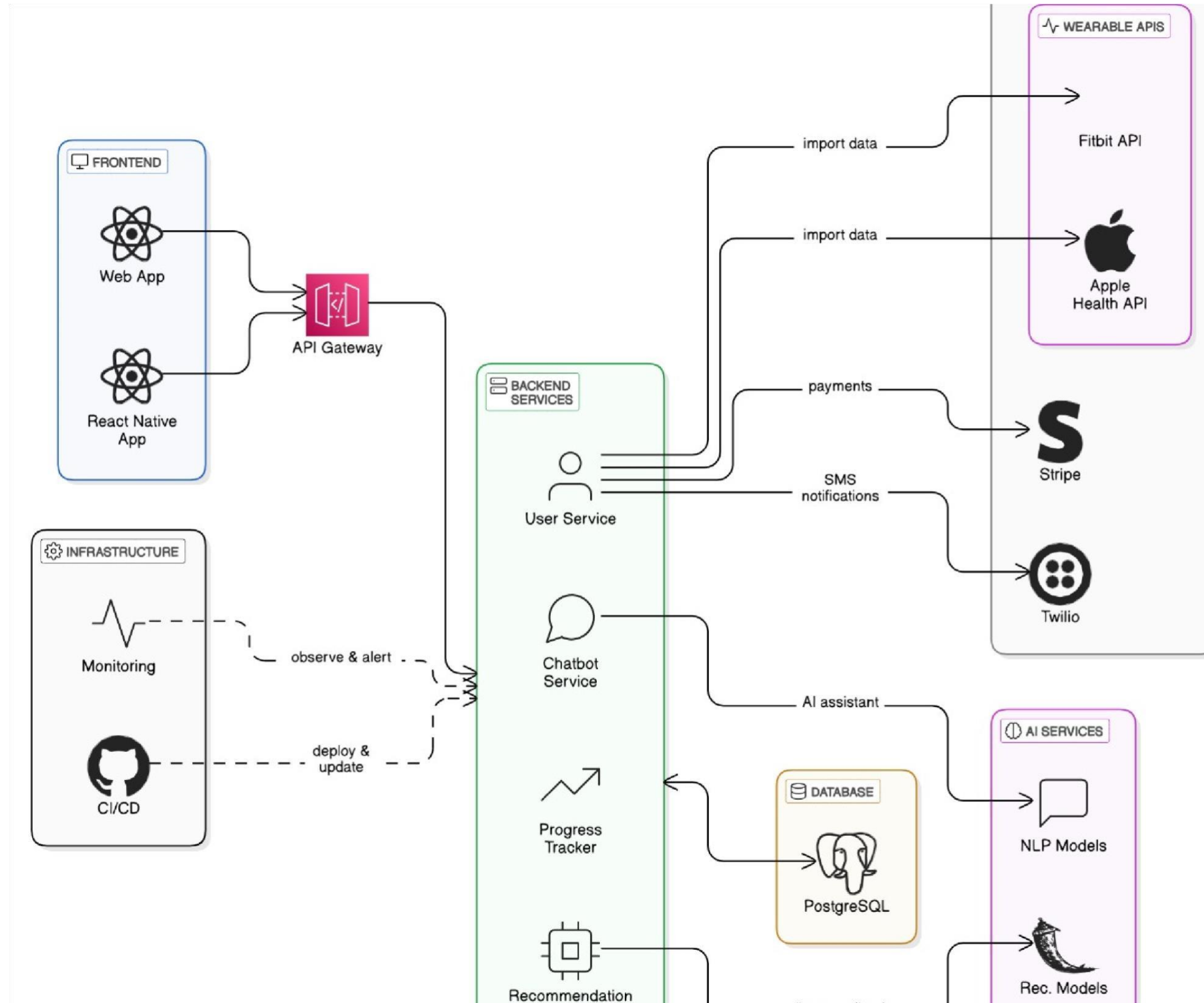
SYSTEM ARCHITECTURE



MLRIT

Raising Engineers

MLR INSTITUTE OF TECHNOLOGY

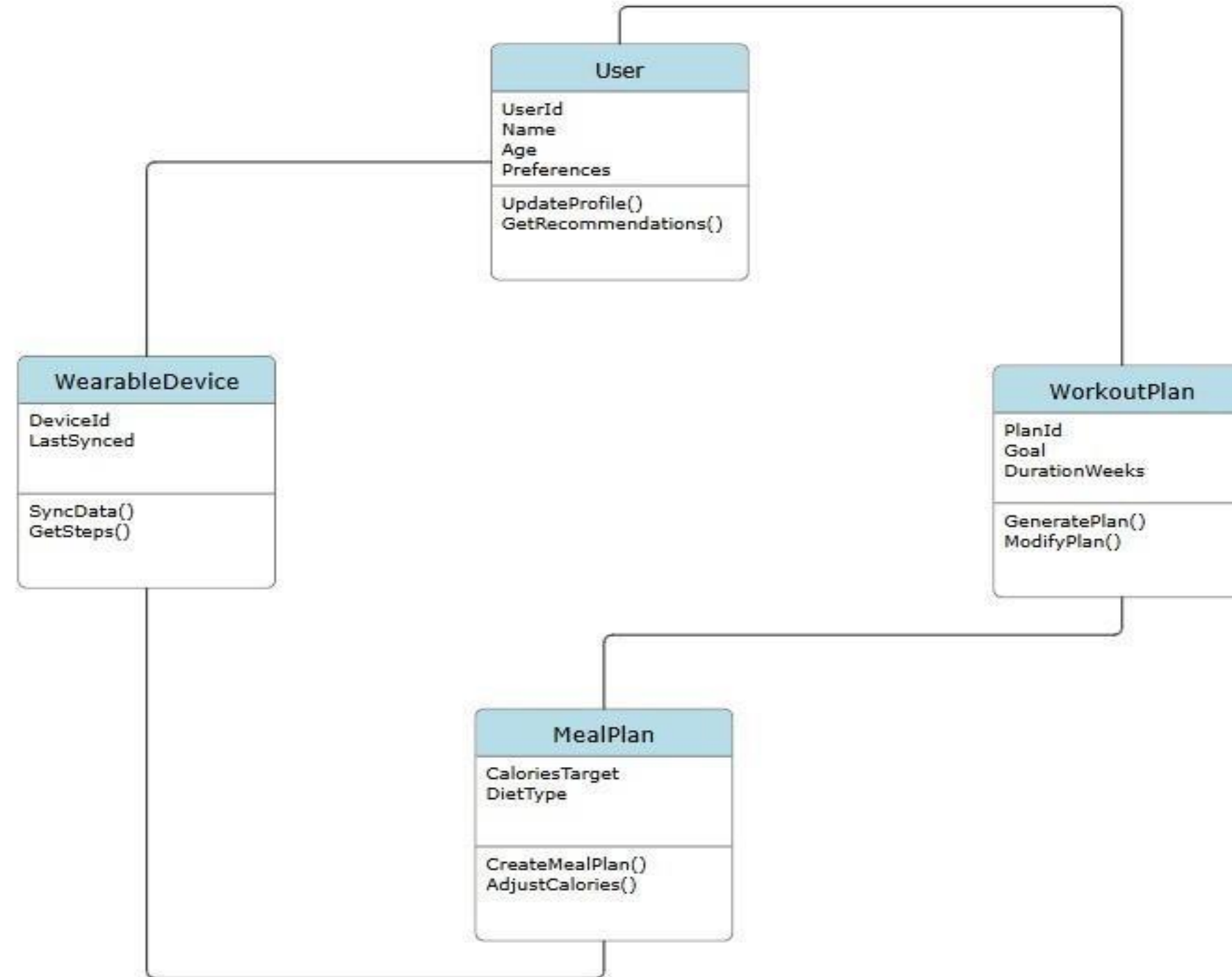


CLASSDIAGRAM

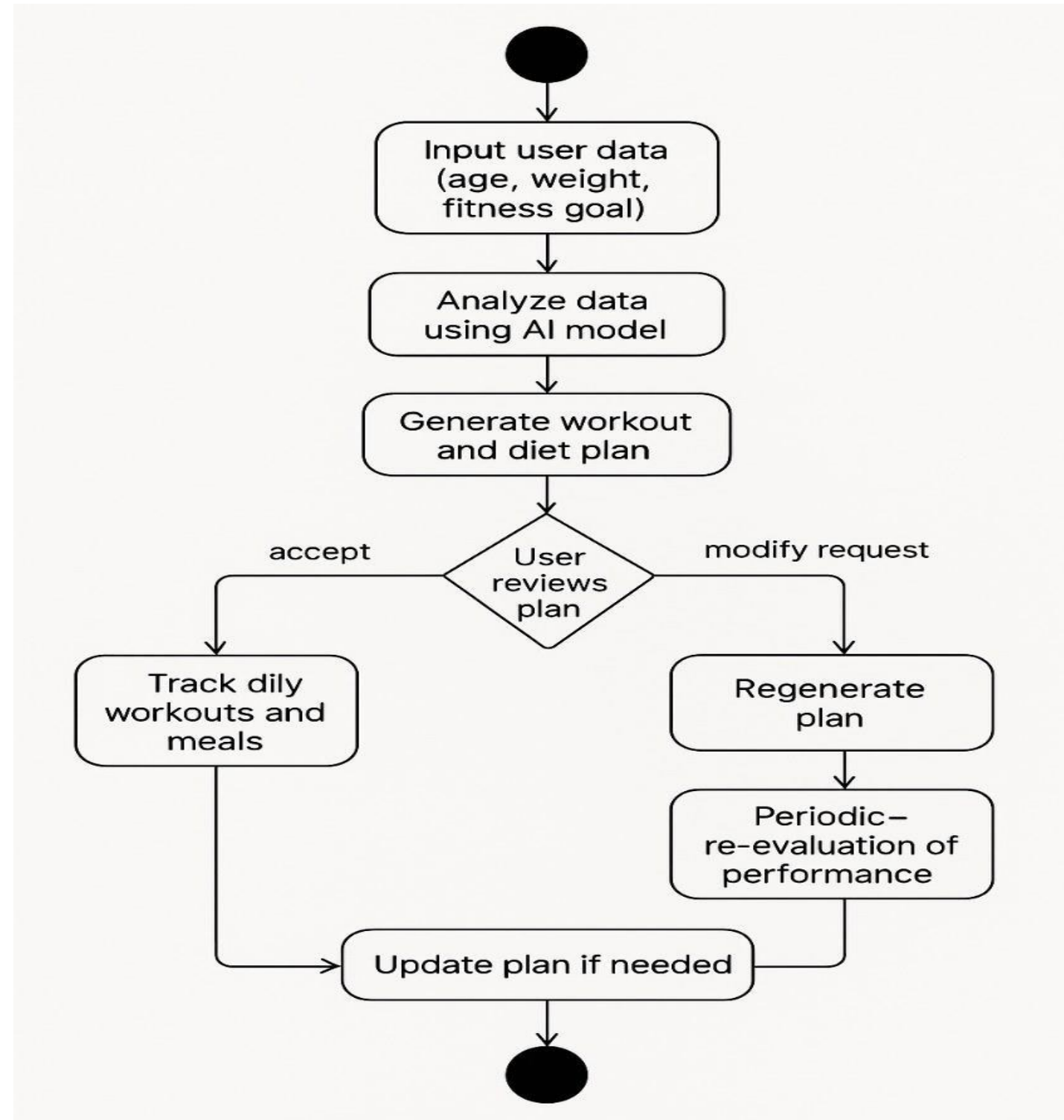


MLRIT
Raising Engineers

MLR INSTITUTE OF TECHNOLOGY



ACTIVITY DIGRAM



• **WHAT IS HYPERFIT AI ?**

- Advanced AI systems optimized for peak performance and efficiency.
- Designed for specific tasks, industries or real-time applications
- combines cutting-edge algorithms, adaptability, and scalability.

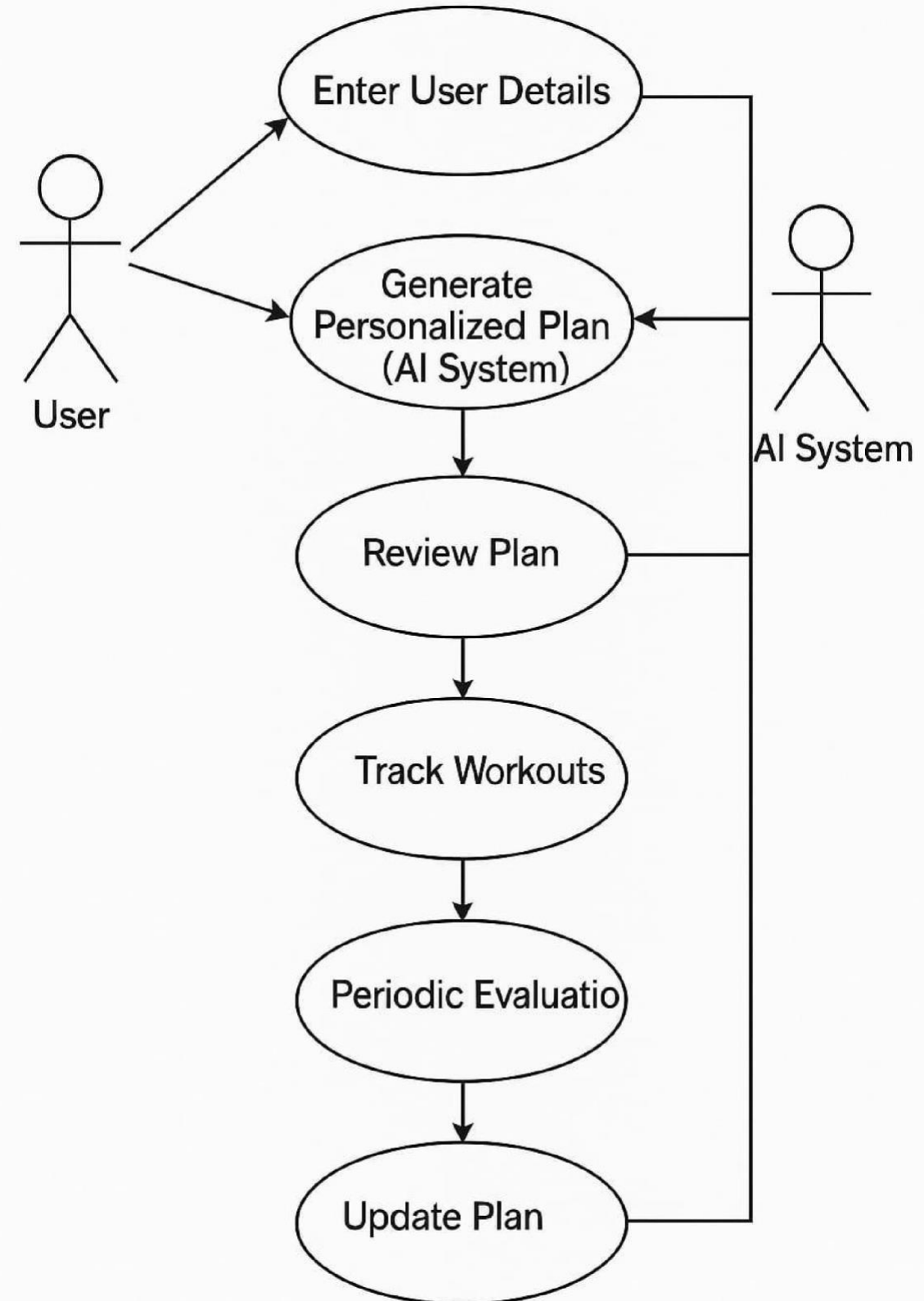


USE CASE DIAGRAM

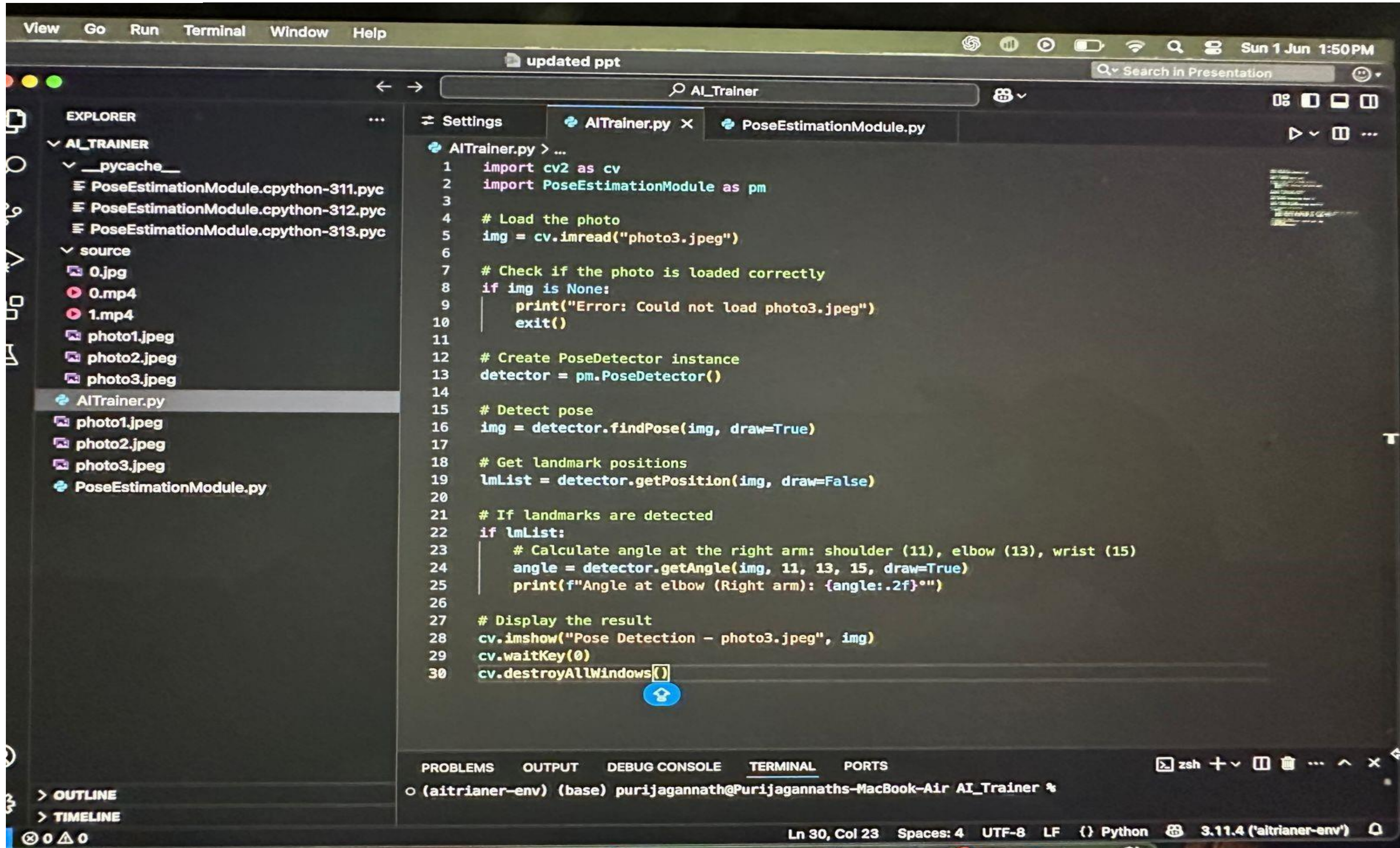
• **WHAT IS HYPERFIT AI ?**

- Advanced AI systems optimized for peak performance and efficiency.
- Designed for specific tasks, industries or real-time applications
- combines cutting-edge algorithms, adaptability, and scalability.





Sample Code



The screenshot shows a Visual Studio Code editor window with a dark theme. The Explorer panel on the left displays a project structure for 'AI_TRAINER' with subfolders '__pycache__' and 'source'. The 'source' folder contains files '0.jpg', '0.mp4', '1.mp4', 'photo1.jpeg', 'photo2.jpeg', 'photo3.jpeg', 'AITrainer.py', and 'PoseEstimationModule.py'. The AITrainer.py file is selected and its code is visible in the main editor. The code imports cv2 and PoseEstimationModule, loads a photo, checks if it's loaded correctly, creates a PoseDetector instance, detects a pose, gets landmark positions, calculates an angle at the elbow, and displays the result. The terminal at the bottom shows the command prompt for the 'aitrianer-env' environment.

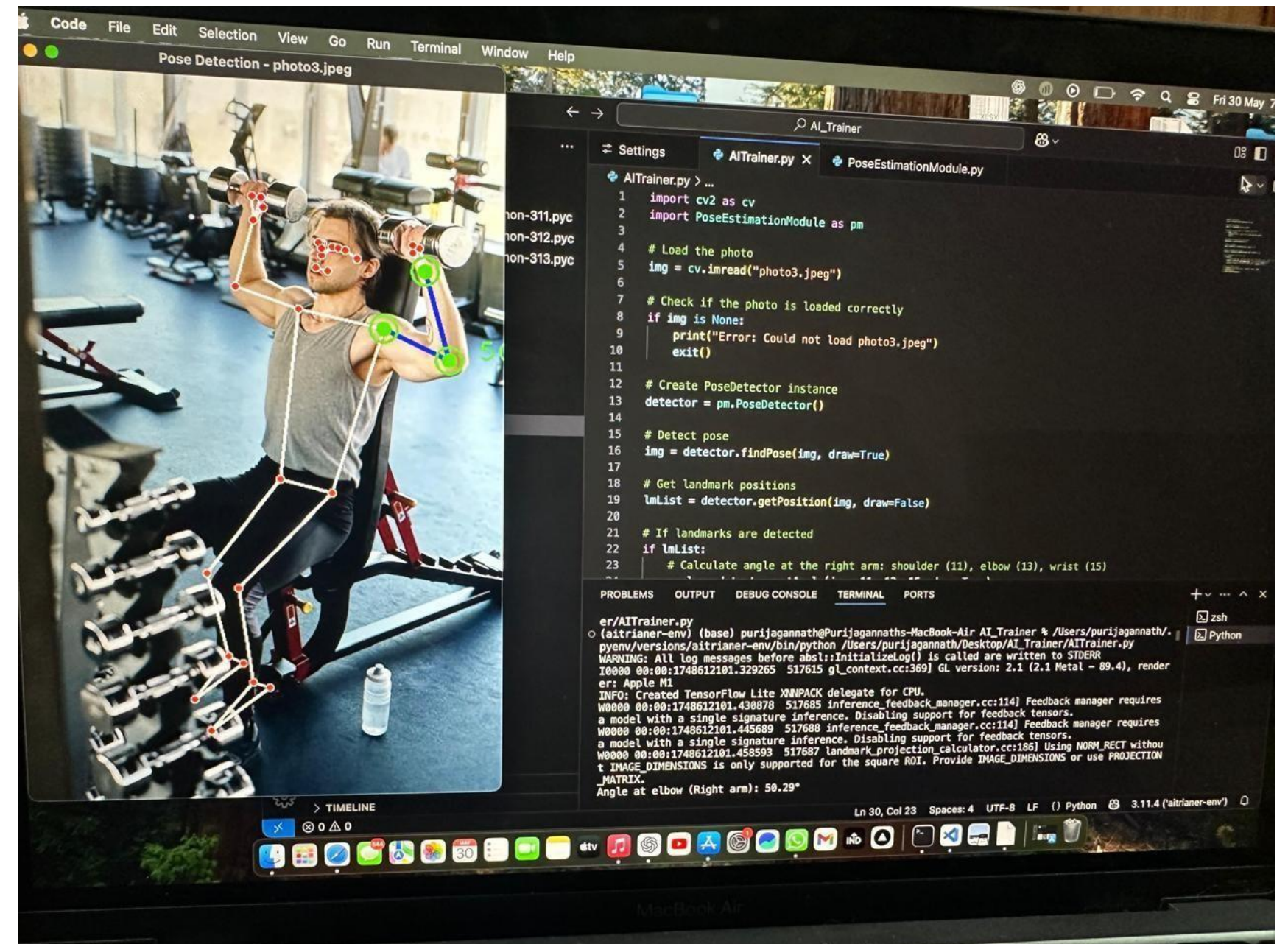
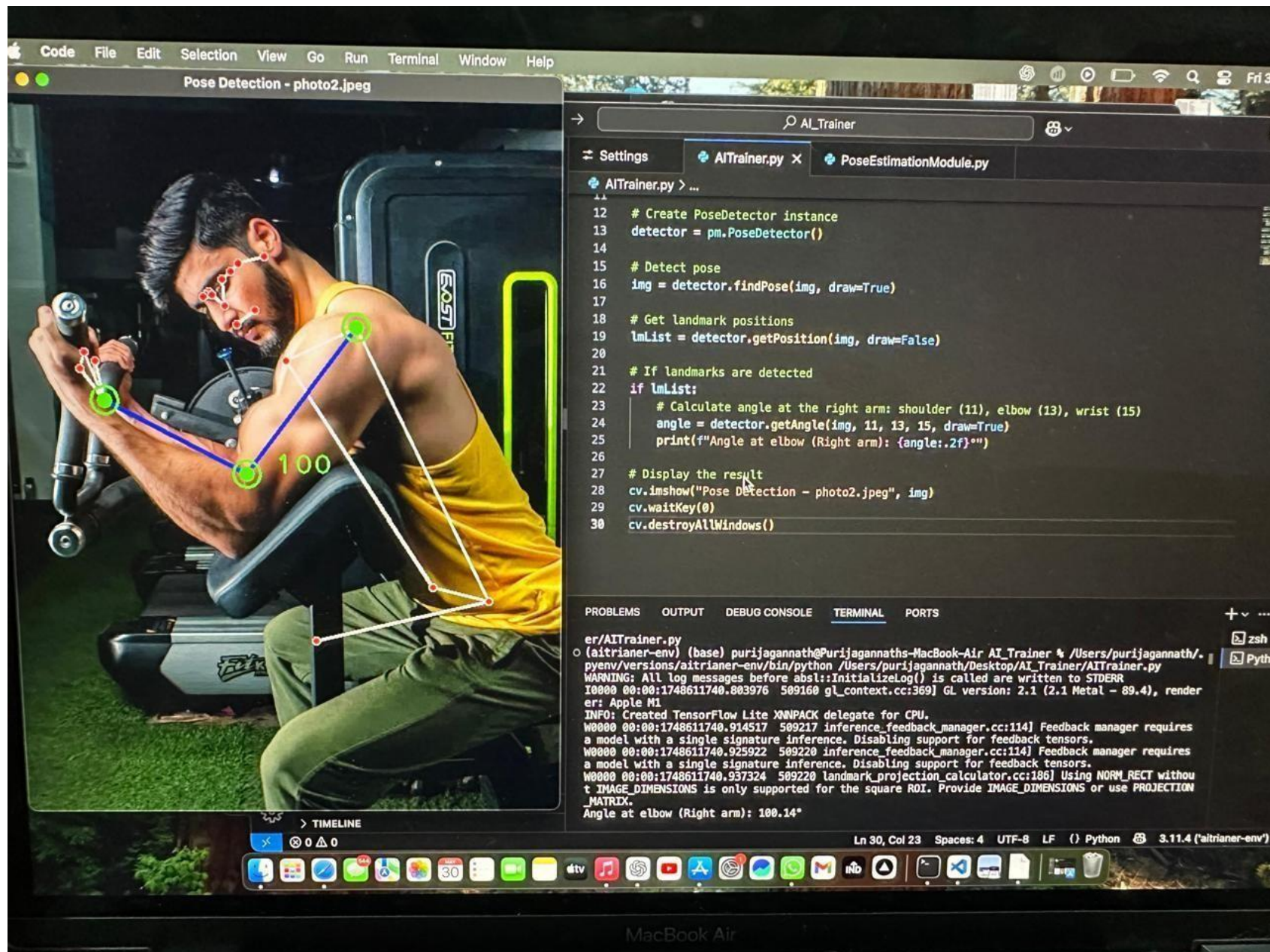
```
1 import cv2 as cv
2 import PoseEstimationModule as pm
3
4 # Load the photo
5 img = cv.imread("photo3.jpeg")
6
7 # Check if the photo is loaded correctly
8 if img is None:
9     print("Error: Could not load photo3.jpeg")
10    exit()
11
12 # Create PoseDetector instance
13 detector = pm.PoseDetector()
14
15 # Detect pose
16 img = detector.findPose(img, draw=True)
17
18 # Get landmark positions
19 lmList = detector.getPosition(img, draw=False)
20
21 # If landmarks are detected
22 if lmList:
23     # Calculate angle at the right arm: shoulder (11), elbow (13), wrist (15)
24     angle = detector.getAngle(img, 11, 13, 15, draw=True)
25     print(f"Angle at elbow (Right arm): {angle:.2f}°")
26
27 # Display the result
28 cv.imshow("Pose Detection - photo3.jpeg", img)
29 cv.waitKey(0)
30 cv.destroyAllWindows()
```

PROBLEMS OUTPUT DEBUG CONSOLE **TERMINAL** PORTS

o (aitrianer-env) (base) purijagannath@Purijagannaths-MacBook-Air AI_Trainer %

Ln 30, Col 23 Spaces: 4 UTF-8 LF {} Python 3.11.4 ('aitrianer-env')

OUTPUTS



Conclusion

• WHAT IS HYPERFIT AI ?

- Advanced AI systems optimized for peak performance and efficiency.
- Designed for specific tasks, industries or real-time applications
- combines cutting-edge algorithms, adaptability, and scalability.



Hyper Fit AI represents the next generation of AI systems, offering tailored, high-performance solutions. Its efficiency, adaptability, and scalability make it a game-changer across industries. As technology continues to evolve, HyperFit AI stands at the forefront of smart, data-driven wellness solutions, bridging the gap between innovation and personalized health experiences.

Future Enhancements

- **Enhance Visualization with a Body Map Module**
Integrate a posture heatmap module that visualizes pressure points and joint alignment. This allows users to see posture deviations in real-time and supports targeted posture correction feedback.
- **Real-Time Alerts for Postural Deviations**

• WHAT IS HYPERFIT AI ?

- Advanced AI systems optimized for peak performance and efficiency.
- Designed for specific tasks, industries or real-time applications
- combines cutting-edge algorithms, adaptability, and scalability.



Enable a real-time alert system that notifies users when posture falls outside optimal alignment based on personalized thresholds. This encourages immediate corrective actions and reduces long-term strain.

- **Activity-Based Posture Analysis for Smarter Feedback**

Implement activity-aware posture categorization (e.g., sitting, standing, exercising) to enable more accurate assessments and context-specific suggestions for improvement

• WHAT IS HYPERFIT AI ?

- Advanced AI systems optimized for peak performance and efficiency.
- Designed for specific tasks, industries or real-time applications
- combines cutting-edge algorithms, adaptability, and scalability.



REFERENCES

1. Jain, A., Sharma, S., & Agrawal, R. (2020). AI-Based Fitness Tracking and Personalization Systems. International Journal of Computer Applications.

<https://www.semanticscholar.org/paper/Machine-Learning-Applications-for-Precision-A-Sharma-Jain/ec9da3b5c896b67a15af062f8bca850526f2e633>

2. Li, T., & Zhang, Y. (2021). Wearable Technology in Fitness and Health Monitoring: A Review. IEEE Access.

https://www.researchgate.net/publication/382878497_Integrating_Wearable_Health_Monitoring_Devices_with_IoT_for_Enhanced_Personal_Health_Management_A_Comprehensive_Review

THANK YOU!